DITTO: A LOW-COST SCREENSHOT INDEX FOR ON-DEVICE EPISODIC RECALL

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ABSTRACT

This document provides a basic paper template and submission guidelines. Abstracts must be a single paragraph, ideally between 4–6 sentences long. Gross violations will trigger corrections at the camera-ready phase.

1 INTRODUCTION

In 1945, Vannevar Bush imagined the memex, a desk in which a person stores all of their books, records, and communications and consults them with speed and flexibility (Bush, 1945). Today's laptops record far more than Bush imagined, but most of it is seen once and then forgotten. People often need to find something they did on their computer days or weeks earlier, such as how they set only one section of a document to landscape, or the monthly rent cap they used when searching for apartments, and finding it again by hand means reopening applications and repeating searches (Hu & Lee, 2020). Screen-recall systems take a step toward the memex: they capture a screenshot every few seconds so that a person can later ask about the past. We call this task *episodic recall* over a person's screen history: given a question, a system returns the screenshots that answer it. Computer-use agents that act on a person's behalf need the same ability (Yang et al., 2026).

Existing screen-recall systems store screenshots but search only the text on them. Screenpipe, OpenRecall, and Rem index text from OCR or the accessibility tree, as did Rewind [cite: Screenpipe, OpenRecall, Rem, Rewind]; Microsoft Recall also indexes screenshots, but it is closed source and requires a 40-TOPS NPU [cite: Microsoft Recall]. Text indexing suits the privacy needs of this task. A screen history records sensitive information, such as the two medications a person checked against each other or the taxable income they entered into a tax calculator, so it should stay on the device, where OCR text is cheap to extract, store, and embed. Systems that send every screenshot to a cloud VLM (Wang & Chen, 2025) do not meet this requirement. However, OCR text keeps the words on a screen and drops the rest: the layout, the icons, and the state of menus and dialogs. The answer to "how did I set one section to landscape?" lies in exactly such menus and dialogs. Is the text on a screen enough to find it again, or should the system search the screenshot itself?

Table 1. Retrieval accuracy and encoding cost of the OCR-text index and the screenshot index on PSAI-A (63,538 screenshots, 5,507 test questions). Both indexes use Qwen3-VL-Embedding-2B with the same LoRA fine-tuning. Encoding time (Sec.) is measured per screenshot on the GPU of an Apple M2 Pro laptop. Min/h is the GPU time in minutes spent on indexing per hour of screen use (720 screenshots). TFLOP is per screenshot. TP: token pruning; KV: K/V reuse in the vision encoder (V) and the language model (L).

Index	R@1	R@10	TFLOP	Sec.	Min/h
OCR text	66.1	79.4	1.22	1.27	15.2
Screenshot	68.6	85.8	6.17	2.86	34.3
+ TP	68.8	84.6	3.12	1.13	13.6
+ TP + KV (V)	68.4	84.3	2.30	1.08	13.0
+ TP + KV (V+L)	68.1	83.9	1.93	0.98	11.8

Screenshots versus OCR text: accuracy and cost. In this paper, we first show that searching the screenshot finds the right moment more often than searching its text. We hold the embedding model fixed (Qwen3-VL-Embedding-2B [cite: Qwen3-VL-Embedding]) and build both indexes over 63,538 screenshots from one person's computer use, with 5,507 test questions about later periods. Without training, the screenshot index outperforms the OCR-text index by 6.1 points in R@1 (65.6% vs. 59.5%). After the same LoRA fine-tuning, Table 1 shows a gap of 2.5 points in R@1 and 6.4 points in R@10. This extends the finding of visual document retrieval (Ma et al., 2024; Faysse et al., 2025) from static documents to screen histories. The screenshot, however, is expensive to encode. Table 1 also shows that encoding one screenshot takes 2.86 s on a laptop GPU, compared with 1.27 s for OCR and text embedding. A recall system encodes every screenshot it captures, so the GPU

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spends 34 minutes of every hour of screen use on indexing, more than twice the 15 minutes of the OCR-text index. Must a laptop then choose between the accuracy of screenshots and the cost of text?

To answer this question, we look at what the encoder spends its compute on. Most of each screenshot repeats what the encoder has already seen. Within a screenshot, large areas such as backgrounds, margins, and whitespace are blank or flat, so each of their blocks can be predicted from its neighbors (spatial redundancy). Across screenshots taken 5 s apart, most of the screen does not change, and when it does, only small regions such as typed text or the cursor differ (temporal redundancy). A screenshot index that encodes every pixel of every screenshot pays for this repetition in full. So, if most of a screenshot repeats its neighbors or the previous screenshot, how much of the encoding can we skip without losing the accuracy of screenshots?

Our solution: Ditto. We present Ditto, a screenshot index that encodes only what does not repeat. The name comes from the ditto mark, which means "the same as above": Ditto does not encode a block that repeats its neighbors or the same block in the previous screenshot. Ditto removes spatial redundancy with token pruning [cite: PixelPrune], which drops each 32×32 block whose pixels can be predicted from its left and upper neighbors before the block enters the vision encoder. It removes temporal redundancy with K/V reuse: a block counts as changed if any of its pixels differs by more than $8/255$ from when it was last encoded, the vision encoder recomputes keys and values only for changed blocks, and the language model reuses the cached keys and values of unchanged visual tokens. Reused keys and values can differ from recomputed ones, so Ditto encodes the whole screenshot again when more than 60% of its blocks change or every 10 screenshots. Neither technique requires training.

We implement Ditto inside Qwen3-VL-Embedding-2B and run it on the GPU of an Apple M2 Pro laptop. Table 1 shows that Ditto reduces encoding compute to 31% of the full screenshot and encoding time to 0.98 s per screenshot, below the 1.27 s of the OCR-text index; indexing then takes 12 GPU minutes per hour of screen use instead of 34. The OCR-text index still uses fewer FLOPs (1.22 vs. 1.93 TFLOP); its text extraction runs on the CPU, so it is slower despite needing less computation. Ditto lowers R@1 by at most 0.5 points and R@10 by at most 1.9 points, and it still outperforms the OCR-text index by 2.0 points in R@1 and 4.5 points in R@10, with 95% confidence intervals that exclude zero. On a laptop GPU, then, the choice between screenshots and text is not forced: screenshots can be encoded faster than OCR text while keeping their accuracy advantage.

Lower encoding cost also changes what a recall system can do. First, a person's history keeps growing, and recall gets harder as it does: in a pilot study with 500 fixed questions, R@1 of the screenshot index drops from 78.8% to 53.2% as the pool grows from 5K to 80K screenshots, while staying ahead of the OCR-text index at every pool size. Ditto makes larger pools affordable to index [TODO: larger-pool results in the main setting]. Second, because Ditto returns screenshots rather than extracted text, an agent can use it to recall a value from the person's past and finish a task, such as finding a yoga mat with the same filters the person used in August [TODO: agentic results]. Thus, the screenshot itself, not its text, can serve as the record that a personal memory system stores and searches, at a cost a laptop can sustain.

To summarize, we make the following contributions:

- We show that, with the same embedding model and training, searching screenshots retrieves better than searching their OCR text on real screen histories, but takes 2.3× longer to encode on a laptop GPU.
- We present Ditto, which removes spatial and temporal redundancy in screen streams with token pruning and K/V reuse, without training. Ditto encodes a screenshot faster than the OCR-text index on a laptop GPU and keeps a lead of 2.0 points in R@1 and 4.5 points in R@10.
- We build an episodic-recall benchmark from the screen recordings of two people (PSAI-A: 117 days, 91.8 hours; PSAI-B: 57 days, 17.2 hours) and from computer-use rollouts (MyPCBench (Jang et al., 2026)), with single-hop, multi-hop, and agentic questions. Each question is checked against the person's whole history, so every recording that answers it counts as gold [TODO: final question counts].

2 Electronic Submission

Submission to MLSys 2025 will be entirely electronic, via a web site (not email). Information about the submission process and LaTeX templates are available on the conference web site at:

https://www.mlsys.org/

The guidelines below will be enforced for initial submissions and camera-ready copies. Here is a brief summary:

- Submissions must be in PDF.
- The maximum paper length is **10 pages** excluding references and appendices. (pages 11 and on must contain only references).

- **Do not include author information or acknowledgements** in your initial submission.
- Your paper should be in **10 point Times font**.
- Make sure your PDF file only uses Type-1 fonts.
- Place figure captions *under* the figure (and omit titles from inside the graphic file itself). Place table captions *over* the table.
- References must include page numbers whenever possible and be as complete as possible. Place multiple citations in chronological order.
- Do not alter the style template; in particular, do not compress the paper format by reducing the vertical spaces.
- Keep your abstract brief and self-contained, one paragraph and roughly 4–6 sentences. Gross violations will require correction at the camera-ready phase. The title should have content words capitalized.
- Submissions that significantly diverge from the specified format will be rejected without review.

2.1 Submitting Papers

Paper Deadline: The deadline for paper submission that is advertised on the conference website is strict. If your full, anonymized, submission does not reach us on time, it will not be considered for publication. There is no separate abstract submission.

Anonymous Submission: MLSys uses double-blind review: no identifying author information may appear on the title page or in the paper itself. Section 3.3 gives further details.

Simultaneous Submission: MLSys will not accept any paper which, at the time of submission, is under review for another conference or has already been published. This policy also applies to papers that overlap substantially in technical content with conference papers under review or previously published. MLSys submissions must not be submitted to other conferences during MLSys's review period. Authors may submit to MLSys substantially different versions of journal papers that are currently under review by the journal, but not yet accepted at the time of submission. Informal publications, such as technical reports or papers in workshop proceedings which do not appear in print, do not fall under these restrictions.

Authors must provide their manuscripts in **PDF** format. Furthermore, please make sure that files contain only embedded Type-1 fonts (e.g., using the program `pdffonts` in linux or using File/DocumentProperties/Fonts in Acrobat). Other fonts (like Type-3) might come from graphics files imported into the document.

Authors using **Word** must convert their document to PDF. Most of the latest versions of Word have the facility to do this automatically. Submissions will not be accepted in Word format or any format other than PDF.

Those who use **LaTeX** should avoid including Type-3 fonts. Those using `latex` and `dvips` may need the following two commands:

```
dvips -Ppdf -tletter -G0 -o paper.ps paper.dvi
ps2pdf paper.ps
```

It is a zero following the "-G", which tells dvips to use the config.pdf file. Newer TeX distributions don't always need this option.

Using `pdflatex` rather than `latex`, often gives better results. This program avoids the Type-3 font problem, and supports more advanced features in the `microtype` package.

Graphics files should be a reasonable size, and included from an appropriate format. Use vector formats (.eps/.pdf) for plots, lossless bitmap formats (.png) for raster graphics with sharp lines, and jpeg for photo-like images.

The style file uses the `hyperref` package to make clickable links in documents. If this causes problems for you, add `nohyperref` as one of the options to the `mlsys2025` usepackage statement.

2.2 Submitting Final Camera-Ready Copy

The final versions of papers accepted for publication should follow the same format and naming convention as initial submissions, except that author information (names and affiliations) should be given. See Section 3.3.2 for formatting instructions.

The footnote, "Preliminary work. Under review by the MLSys. Do not distribute." must be modified to "*Proceedings of the 8th MLSys Conference*, Santa Clara, CA, USA, 2025. Copyright 2025 by the author(s)."

For those using the **LaTeX** style file, this change (and others) is handled automatically by simply changing `\usepackage{mlsys2025}` to

`\usepackage[accepted]{mlsys2025}`

Authors using **Word** must edit the footnote on the first page of the document themselves.

Camera-ready copies should have the title of the paper as running head on each page except the first one. The running title consists of a single line centered above a horizontal rule which is 1 point thick. The running head should be centered,

bold and in 9 point type. The rule should be 10 points above the main text. For those using the LaTeX style file, the original title is automatically set as running head using the `fancyhdr` package which is included in the MLSys 2025 style file package. In case that the original title exceeds the size restrictions, a shorter form can be supplied by using

```
\mlsystitlerunning{...}
```

just before `\begin{document}`. Authors using **Word** must edit the header of the document themselves.

3 Format of the Paper

All submissions must follow the specified format.

3.1 Length and Dimensions

Papers must not exceed ten (10) pages, including all figures, tables, but excluding acknowledgements, references and appendices. Acknowledgements should be limited to grants and people who contributed to the paper. Any submission that exceeds this page limit, or that diverges significantly from the specified format, will be rejected without review.

The text of the paper should be formatted in two columns, with an overall width of 6.75 inches, height of 9.0 inches, and 0.25 inches between the columns. The left margin should be 0.75 inches and the top margin 1.0 inch (2.54 cm). The right and bottom margins will depend on whether you print on US letter or A4 paper, but all final versions must be produced for US letter size.

The paper body should be set in 10 point type with a vertical spacing of 11 points. Please use Times typeface throughout the text.

3.2 Title

The paper title should be set in 14 point bold type and centered between two horizontal rules that are 1 point thick, with 1.0 inch between the top rule and the top edge of the page. Capitalize the first letter of content words and put the rest of the title in lower case.

3.3 Author Information for Submission

MLSys uses double-blind review, so author information must not appear. If you are using LaTeX and the `mlsys2025.sty` file, use `\mlsysauthor{...}` to specify authors and `\mlsysaffiliation{...}` to specify affiliations. (Read the TeX code used to produce this document for an example usage.) The author information will not be printed unless `accepted` is passed as an argument to the style file. Submissions that include the author information will not be reviewed.

3.3.1 Self-Citations

If you are citing published papers for which you are an author, refer to yourself in the third person. In particular, do not use phrases that reveal your identity (e.g., "in previous work (Langley, 2000), we have shown ...").

Do not anonymize citations in the reference section. The only exception are manuscripts that are not yet published (e.g., under submission). If you choose to refer to such unpublished manuscripts (Author, 2018), anonymized copies have to be submitted as Supplementary Material via CMT. However, keep in mind that an MLSys paper should be self contained and should contain sufficient detail for the reviewers to evaluate the work. In particular, reviewers are not required to look at the Supplementary Material when writing their review.

3.3.2 Camera-Ready Author Information

If a paper is accepted, a final camera-ready copy must be prepared. For camera-ready papers, author information should start 0.3 inches below the bottom rule surrounding the title. The authors' names should appear in 10 point bold type, in a row, separated by white space, and centered. Author names should not be broken across lines. Unbolded superscripted numbers, starting 1, should be used to refer to affiliations.

Affiliations should be numbered in the order of appearance. A single footnote block of text should be used to list all the affiliations.

Each distinct affiliations should be listed once. If an author has multiple affiliations, multiple superscripts should be placed after the name, separated by thin spaces. If the authors would like to highlight equal contribution by multiple first authors, those authors should have an asterisk placed after their name in superscript, and the term "*Equal contribution" should be placed in the footnote block ahead of the list of affiliations. A list of corresponding authors and their emails (in the format Full Name <email@domain.com>) can follow the list of affiliations. Ideally only one or two names should be listed.

A sample file with author names is included in the mlsys2025 style file package. Turn on the `[accepted]` option to the stylefile to see the names rendered. All of the guidelines above are implemented by the LaTeX style file.

3.4 Abstract

The paper abstract should begin in the left column, 0.4 inches below the final address. The heading 'Abstract' should be centered, bold, and in 11 point type. The abstract body should use 10 point type, with a vertical spacing of 11 points, and should be indented 0.25 inches more than nor-

mal on left-hand and right-hand margins. Insert 0.4 inches of blank space after the body. Keep your abstract brief and self-contained, limiting it to one paragraph and roughly 4–6 sentences. Gross violations will require correction at the camera-ready phase.

3.5 Partitioning the Text

You should organize your paper into sections and paragraphs to help readers place a structure on the material and understand its contributions.

3.5.1 Sections and Subsections

Section headings should be numbered, flush left, and set in 11 pt bold type with the content words capitalized. Leave 0.25 inches of space before the heading and 0.15 inches after the heading.

Similarly, subsection headings should be numbered, flush left, and set in 10 pt bold type with the content words capitalized. Leave 0.2 inches of space before the heading and 0.13 inches afterward.

Finally, subsubsection headings should be numbered, flush left, and set in 10 pt small caps with the content words capitalized. Leave 0.18 inches of space before the heading and 0.1 inches after the heading.

Please use no more than three levels of headings.

3.5.2 Paragraphs and Footnotes

Within each section or subsection, you should further partition the paper into paragraphs. Do not indent the first line of a given paragraph, but insert a blank line between succeeding ones.

You can use footnotes[1] to provide readers with additional information about a topic without interrupting the flow of the paper. Indicate footnotes with a number in the text where the point is most relevant. Place the footnote in 9 point type at the bottom of the column in which it appears. Precede the first footnote in a column with a horizontal rule of 0.8 inches.[2]

3.6 Figures

You may want to include figures in the paper to illustrate your approach and results. Such artwork should be centered, legible, and separated from the text. Lines should be dark and at least 0.5 points thick for purposes of reproduction, and text should not appear on a gray background.

[1]Footnotes should be complete sentences.

[2]Multiple footnotes can appear in each column, in the same order as they appear in the text, but spread them across columns and pages if possible.

Algorithm 1 Bubble Sort

Input: data x_i, size m
repeat
  Initialize $noChange = true$.
  for $i = 1$ **to** $m - 1$ **do**
    if $x_i > x_{i+1}$ **then**
      Swap x_i and x_{i+1}
      $noChange = false$
    end if
  end for
until $noChange$ is $true$

Label all distinct components of each figure. If the figure takes the form of a graph, then give a name for each axis and include a legend that briefly describes each curve. Do not include a title inside the figure; instead, the caption should serve this function.

Number figures sequentially, placing the figure number and caption *after* the graphics, with at least 0.1 inches of space before the caption and 0.1 inches after it. The figure caption should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left. You may float figures to the top or bottom of a column, and you may set wide figures across both columns (use the environment `figure*` in LaTeX). Always place two-column figures at the top or bottom of the page.

3.7 Algorithms

If you are using LaTeX, please use the "algorithm" and "algorithmic" environments to format pseudocode. These require the corresponding stylefiles, algorithm.sty and algorithmic.sty, which are supplied with this package. Algorithm 1 shows an example.

3.8 Tables

You may also want to include tables that summarize material. Like figures, these should be centered, legible, and numbered consecutively. However, place the title *above* the table with at least 0.1 inches of space before the title and the same after it, as in Table 2. The table title should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left.

Tables contain textual material, whereas figures contain graphical material. Specify the contents of each row and column in the table's topmost row. Again, you may float tables to a column's top or bottom, and set wide tables across both columns. Place two-column tables at the top or bottom of the page.

Table 2. Classification accuracies for naive Bayes and flexible Bayes on various data sets.

DATA SET	NAIVE	FLEXIBLE	BETTER?
BREAST	95.9± 0.2	96.7± 0.2	√
CLEVELAND	83.3± 0.6	80.0± 0.6	×
GLASS2	61.9± 1.4	83.8± 0.7	√
CREDIT	74.8± 0.5	78.3± 0.6	
HORSE	73.3± 0.9	69.7± 1.0	×
META	67.1± 0.6	76.5± 0.5	√
PIMA	75.1± 0.6	73.9± 0.5	
VEHICLE	44.9± 0.6	61.5± 0.4	√

3.9 Citations and References

Please use APA reference format regardless of your formatter or word processor. If you rely on the LaTeX bibliographic facility, use `natbib.sty` and `mlsys2025.bst` included in the style-file package to obtain this format.

Citations within the text should include the authors' last names and year. If the authors' names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel's pioneering work (1959). Otherwise place the entire reference in parentheses with the authors and year separated by a comma (Samuel, 1959). List multiple references separated by semicolons (Kearns, 1989; Samuel, 1959; Mitchell, 1980). Use the 'et al.' construct only for citations with three or more authors or after listing all authors to a publication in an earlier reference (Michalski et al., 1983).

Authors should cite their own work in the third person in the initial version of their paper submitted for blind review. Please refer to Section 3.3 for detailed instructions on how to cite your own papers.

Use an unnumbered first-level section heading for the references, and use a hanging indent style, with the first line of the reference flush against the left margin and subsequent lines indented by 10 points. The references at the end of this document give examples for journal articles (Samuel, 1959), conference publications (Langley, 2000), book chapters (Newell & Rosenbloom, 1981), books (Duda et al., 2000), edited volumes (Michalski et al., 1983), technical reports (Mitchell, 1980), and dissertations (Kearns, 1989).

Alphabetize references by the surnames of the first authors, with single author entries preceding multiple author entries. Order references for the same authors by year of publication, with the earliest first. Make sure that each reference includes all relevant information (e.g., page numbers).

Please put some effort into making references complete, presentable, and consistent. If using bibtex, please protect capital letters of names and abbreviations in titles, for example, use {B}ayesian or {L}ipschitz in your .bib file.

3.10 Software and Data

We strongly encourage the publication of software and data with the camera-ready version of the paper whenever appropriate. This can be done by including a URL in the camera-ready copy. However, do not include URLs that reveal your institution or identity in your submission for review. Instead, provide an anonymous URL or upload the material as "Supplementary Material" into the CMT reviewing system. Note that reviewers are not required to look at this material when writing their review.

ACKNOWLEDGEMENTS

Do not include acknowledgements in the initial version of the paper submitted for blind review.

If a paper is accepted, the final camera-ready version can (and probably should) include acknowledgements. In this case, please place such acknowledgements in an unnumbered section at the end of the paper. Typically, this will include thanks to reviewers who gave useful comments, to colleagues who contributed to the ideas, and to funding agencies and corporate sponsors that provided financial support.

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A PLEASE ADD SUPPLEMENTAL MATERIAL AS APPENDIX HERE

Put anything that you might normally include after the references as an appendix here, *not in a separate supplementary file*. Upload your final camera-ready as a single pdf, including all appendices.